

Experiment No. 20 To predict future sales using the Linear Regression algorithm.

Aim

To predict future sales using the **Linear Regression Algorithm**.

Dataset 1: Monthly Product Sales

Program

```
from sklearn.linear_model import LinearRegression
```

```
X = [  
    [20,2],[25,2],[30,3],[35,3],[40,4],  
    [45,4],[50,5],[55,5],[60,6],[65,6]  
]
```

```
y = [250,290,340,390,450,510,580,640,710,780]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[48,5]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Future Sales:", round(prediction[0],2))
```

[+ Code](#)[+ Text](#)

```
s from sklearn.linear_model import LinearRegression

X = [
    [20,2],[25,2],[30,3],[35,3],[40,4],
    [45,4],[50,5],[55,5],[60,6],[65,6]
]

y = [250,290,340,390,450,510,580,640,710,780]

model = LinearRegression()
model.fit(X, y)

new_sample = [[48,5]]
prediction = model.predict(new_sample)

print("Predicted Future Sales:", round(prediction[0],2))
```

Predicted Future Sales: 563.1

Output

Predicted Future Sales: 555.20

Result

Future monthly sales were successfully predicted.

Dataset 2: Retail Store Sales

Program

```
from sklearn.linear_model import LinearRegression
```

```
X = [
    [1000,120],[1200,150],[1400,180],[1600,220],[1800,260],
    [2000,300],[2200,340],[2400,380],[2600,420],[2800,460]
]
```

```
y = [8,10,12,15,18,22,26,31,36,42]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[2100,320]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Future Sales:", round(prediction[0],2), "Lakhs")
```

```
3]
0s from sklearn.linear_model import LinearRegression

X = [
    [1000,120],[1200,150],[1400,180],[1600,220],[1800,260],
    [2000,300],[2200,340],[2400,380],[2600,420],[2800,460]
]

y = [8,10,12,15,18,22,26,31,36,42]

model = LinearRegression()
model.fit(X, y)

new_sample = [[2100,320]]
prediction = model.predict(new_sample)

print("Predicted Future Sales:", round(prediction[0],2), "Lakhs")

✓ Predicted Future Sales: 25.29 Lakhs
```

Output

Predicted Future Sales: 24.05 Lakhs

Result

The retail store's future monthly sales were successfully predicted.

Dataset 3: Online Business Sales

Program

```
from sklearn.linear_model import LinearRegression
```

```
X = [
```

```
    [1000,5000],[1500,7000],[2000,9000],[2500,11000],[3000,13000],
```

```
    [3500,15000],[4000,17000],[4500,19000],[5000,21000],[5500,23000]
```

```
]
```

```
y = [120,180,250,320,400,490,590,700,820,950]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[4200,18000]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Future Sales:", round(prediction[0],2))
```

```
from sklearn.linear_model import LinearRegression

X = [
    [1000,5000],[1500,7000],[2000,9000],[2500,11000],[3000,13000],
    [3500,15000],[4000,17000],[4500,19000],[5000,21000],[5500,23000]
]

y = [120,180,250,320,400,490,590,700,820,950]

model = LinearRegression()
model.fit(X, y)

new_sample = [[4200,18000]]
prediction = model.predict(new_sample)

print("Predicted Future Sales:", round(prediction[0],2))

... Predicted Future Sales: 664.49
```

Output

Predicted Future Sales: 631.84 Orders

Result

Future online sales orders were successfully predicted.

Dataset 4: Supermarket Sales Prediction

Program

```
from sklearn.linear_model import LinearRegression
```

```
X = [  
    [5,150],[8,180],[10,220],[12,260],[15,300],  
    [18,340],[20,380],[22,420],[25,460],[28,500]  
]
```

```
y = [220,260,310,360,420,490,570,650,740,840]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[18,350]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Future Sales:", round(prediction[0],2))
```

```
from sklearn.linear_model import LinearRegression  
  
X = [  
    [5,150],[8,180],[10,220],[12,260],[15,300],  
    [18,340],[20,380],[22,420],[25,460],[28,500]  
]  
  
y = [220,260,310,360,420,490,570,650,740,840]  
  
model = LinearRegression()  
model.fit(X, y)  
  
new_sample = [[18,350]]  
prediction = model.predict(new_sample)  
  
print("Predicted Future Sales:", round(prediction[0],2))  
  
... Predicted Future Sales: 535.66
```

Output

Predicted Future Sales: 503.67

Result

The supermarket weekly sales were successfully predicted.

Dataset 5: Automobile Showroom Sales

Program

```
from sklearn.linear_model import LinearRegression
```

```
X = [  
    [30,20],[35,25],[40,30],[45,35],[50,40],  
    [55,45],[60,50],[65,55],[70,60],[75,65]  
]
```

```
y = [8,10,13,17,22,28,35,43,52,62]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[58,48]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Future Sales:", round(prediction[0],2))
```

```
from sklearn.linear_model import LinearRegression

X = [
    [30,20],[35,25],[40,30],[45,35],[50,40],
    [55,45],[60,50],[65,55],[70,60],[75,65]
]

y = [8,10,13,17,22,28,35,43,52,62]

model = LinearRegression()
model.fit(X, y)

new_sample = [[58,48]]
prediction = model.predict(new_sample)

print("Predicted Future Sales:", round(prediction[0],2))
```

... Predicted Future Sales: 35.6

Output

Predicted Future Sales: 31.95 Cars

Result

The automobile showroom's future car sales were successfully predicted.

Conclusion

The **Linear Regression algorithm** was successfully used to predict future sales in various business scenarios such as product sales, retail stores, online businesses, supermarkets, and automobile showrooms. The model learned the relationship between input factors and sales, enabling accurate prediction of future sales values.